

Automated Malware Detection and Removal using Wazuh Active Response with VirusTotal

Project Overview

In modern Security Operations Centers (SOC), detecting malware is not sufficient on its own. Security teams must **respond automatically and quickly** to prevent damage, lateral movement, and persistence. Manual incident response introduces delays and increases risk.

This project demonstrates how **Wazuh Active Response**, integrated with **VirusTotal**, can be used to **automatically remove malicious files** from endpoints immediately after detection — without analyst intervention.

The solution uses:

- **Wazuh File Integrity Monitoring (FIM)** for real-time detection
- **VirusTotal** for malware reputation analysis
- **Wazuh Active Response** to execute a remediation script
- **TheHive** for SOC-level visibility and incident tracking

The result is a **fully automated detect → analyze → respond → remediate pipeline**.

Why Active Response with VirusTotal?

CHALLENGE	TRADITIONAL SOC	THIS PROJECT
MALWARE RESPONSE TIME	Manual	Automated
ANALYST WORKLOAD	High	Reduced
ENDPOINT EXPOSURE	Minutes–Hours	Seconds
HUMAN ERROR	Possible	Eliminated
SCALABILITY	Limited	High

Key Benefit:

Malware is **deleted automatically** as soon as VirusTotal confirms it is malicious.

Tools & Technologies Used

- **Wazuh SIEM / XDR**
- **VirusTotal Public API**
- **Linux (Ubuntu Endpoint)**
- **Bash Scripting**
- **TheHive (Incident Response Platform)**

Architecture Overview

Flow:

1. Endpoint file change detected by Wazuh FIM
2. VirusTotal scans the file hash
3. Malicious verdict triggers a Wazuh rule
4. Active Response executes a removal script on the agent
5. File is deleted from disk
6. Logs and alerts are generated for SOC visibility

Components Used:

- Wazuh Manager
- Wazuh Agent (Ubuntu endpoint)
- VirusTotal API
- Bash Active Response Script
- Wazuh Dashboard
- TheHive

❖ Implementation Steps

Step 1: Create Custom Rules for Active Response Results

Purpose:

To generate alerts indicating whether malware removal was **successful or failed**.

Action:

Custom rules were added on the **Wazuh Manager** to parse active response logs and create readable alerts.

Outcome:

- Alert for **Successful removal**
- Alert for **Failed removal**

```
</group>
<group name="virustotal,active_response">
  <rule id="100092" level="12">
    <if_sid>657</if_sid>
    <match>Successfully removed threat</match>
    <description>Active Response removed malware: $(parameters.alert.data.virustotal.source.file)</description>
  </rule>
  <rule id="100093" level="12">
    <if_sid>657</if_sid>
    <match>Error removing threat</match>
    <description>Active Response failed to remove malware: $(parameters.alert.data.virustotal.source.file)</description>
  </rule>
</group>
```

Step 2: Configure Active Response on Wazuh Manager

Purpose:

To instruct Wazuh Manager to trigger an active response when VirusTotal detects malware.

Action:

- Added a <command> block defining the response script
- Added an <active-response> block bound to the VirusTotal malware rule ID

Key Trigger Rule:

87105 (VirusTotal: Malicious file detected)

```
<command>
  <name>remove-threat</name>
  <executable>remove-threat.sh</executable>
  <timeout_allowed>no</timeout_allowed>
</command>

<active-response>
  <disabled>no</disabled>
  <command>remove-threat</command>
  <location>local</location>
  <rules_id>87105</rules_id>
</active-response>
```

Step 3: Create Active Response Script on Wazuh Agent

Purpose:

To remove the malicious file automatically from the endpoint.

Action:

A Bash script (remove-threat.sh) was created on the **agent** to:

- Parse the alert JSON
- Extract the malicious file path
- Delete the file
- Log success or failure

```

GNU nano 7.2 /var/ossec/active-response/bin/remove-threat.sh
#!/bin/bash

LOCAL=`dirname $0`;
cd $LOCAL
cd ../

PWD=`pwd`

read INPUT_JSON
FILENAME=$(echo $INPUT_JSON | jq -r .parameters.alert.data.virustotal.source.file)
COMMAND=$(echo $INPUT_JSON | jq -r .command)
LOG_FILE="$PWD/../logs/active-responses.log"

#----- Analyze command -----#
if [ ${COMMAND} = "add" ]
then
    # Send control message to execd
    printf '{"version":1,"origin":{"name":"remove-threat","module":"active-response"},"command":"check_keys", "parameters":{"keys":[]}}\n'

```

Step 4: Set Script Permissions

Purpose:

Ensure the script can be executed securely by Wazuh.

Action:

- Set execution permissions
- Changed ownership to root:wazuh

```

root@ip- /home/ubuntu# nano /var/ossec/active-response/bin/remove-threat.sh
root@ip- /home/ubuntu# sudo chmod 750 /var/ossec/active-response/bin/remove-threat.sh
sudo chown root:wazuh /var/ossec/active-response/bin/remove-threat.sh

```

Step 5: Malware Simulation Test

Purpose:

Validate the full detection and response workflow.

Action:

- Downloaded the **EICAR test malware** to the monitored directory
- Triggered FIM detection and VirusTotal scan

```

root@ip- :/home/ubuntu# sudo curl -Lo /root/bad.exe https://secure.eicar.org/eicar.com
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100    68    100    68     0     0    200      0 --:--:-- --:--:-- --:--:--    200

```

Step 6: Verify Active Response Execution on Endpoint

Purpose:

Confirm that the file was actually removed.

Action:

- Checked Active Response logs
- Verified file deletion using ls

Result:

✓ Malware file was **successfully removed**

```
root@ip-10-0-1-135: ~$ cat /var/ossec/logs/active-responses.log
Fri Jan 30 17:34:58 IST 2026 active-response/bin/restart.sh agent reload
Sun Feb 1 13:14:31 IST 2026 active-response/bin/restart.sh agent reload
Sun Feb 1 14:40:19 IST 2026 active-response/bin/restart.sh agent reload
2026/02/01 14:50:16 active-response/bin/remove-threat.sh: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"add","parameters":{"extra_args":[],"alert":{"timestamp":"2026-02-01T09:20:16.829+0000","rule":{"level":12,"description":"VirusTotal: Alert - /root/bad.exe - 67 engines detected this file","id":"87105","mitre":{"id":["T1203"],"tactic":["Execution"],"technique":["Exploitation for Client Execution"]},"firedtimes":1,"mail":true,"groups":["virustotal"],"pci_dss":{"10.6.1","11.4"},"gdpr":{"IV_35.7.d"},"agent":{"id":"005","name":"cortex","ip":"10.0.1.144"},"manager":{"name":"ip-10-0-1-135"},"id":"1769937616.2241266","decoder":{"name":"json"},"data":{"virustotal":{"found":"1","malicious":"1","source":{"alert_id":"1769937584.2237371","file":"/root/bad.exe"},"md5":"44d88612fea8a8f36de82e1278abb02f","sha1":"3395856ce81f2b7382dee72602f798b642f14140"},"sha256":"3395856ce81f2b7382dee72602f798b642f14140","scan_date":"2026-02-01 09:10:35","positives":"67","total":"69","permalink":"https://www.virustotal.com/gui/file/275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f/detection/f-275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f-1769937035"},"integration":"virustotal"},"location":"virustotal"},"program":"active-response/bin/remove-threat.sh"}} Successfully removed threat
root@ip-10-0-1-135: ~$ cat /home/ubuntu# ls -lah /root/bad.exe
ls: cannot access '/root/bad.exe': No such file or directory
```

Step 7: Validate Alerts in Wazuh Dashboard

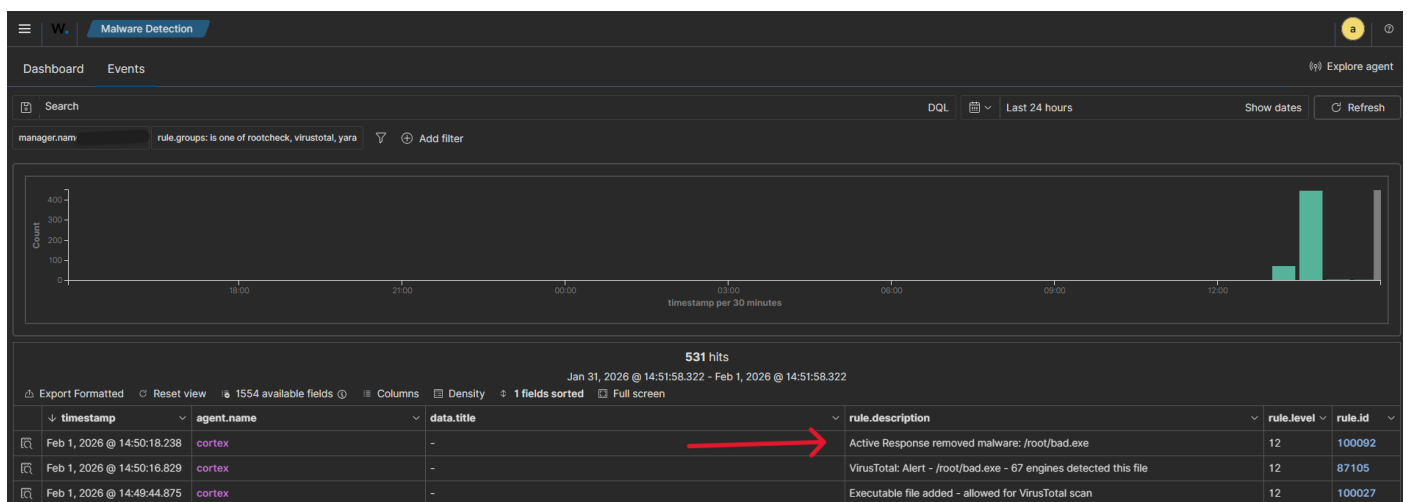
Purpose:

Ensure SOC analysts can clearly see:

- Malware detection
- Automated response execution

Observed Alerts:

1. VirusTotal malware detection alert
2. Active Response executed
3. Malware removed confirmation



t	location	/var/ossec/logs/active-responses.log
t	manager.name	ip-10-0-1-135
t	rule.description	Active Response removed malware: /root/bad.exe
#	rule.firedtimes	1
t	rule.groups	virustotal, active_response
t	rule.id	100092
#	rule.level	12
o	rule.mail	true

⑦	data.parameters.alert.data.v	/root/bad.exe
	irustotal.source.file	
②	data.parameters.alert.data.v	44d88612fea8a8f36de82e1278abb02f
	irustotal.source.md5	
②	data.parameters.alert.data.v	3395856ce81f2b7382dee72602f798b642f14140
	irustotal.source.sha1	
②	data.parameters.alert.data.v	69
	irustotal.total	
t	data.parameters.alert.decode	json
	r.name	
t	data.parameters.alert.id	1769937616.2241266
t	data.parameters.alert.locati	virustotal
	on	
t	data.parameters.alert.manage	
	r.name	
t	data.parameters.alert.rule.d	VirusTotal: Alert - /root/bad.exe - 67 engines detected this file
	escription	
t	data.parameters.alert.rule.f	1
	iredtimes	
⑦	data.parameters.alert.rule.g	IV_35.7.d
	dpr	
t	data.parameters.alert.rule.g	virustotal
	roups	
t	data.parameters.alert.rule.i	87105
	d	
t	data.parameters.alert.rule.l	12
	evel	
t	data.parameters.alert.rule.m	true
	ail	
t	data.parameters.alert.rule.m	T1203
	itre.id	
t	data.parameters.alert.rule.m	Execution
	itre.tactic	
t	data.parameters.alert.rule.m	Exploitation for Client Execution
	itre.technique	
⑦	data.parameters.alert.rule.p	10.6.1, 11.4
	ci_dss	
t	data.parameters.alert.timest	2026-02-01T09:20:16.829+0000
	amp	
t	data.parameters.extra_args	
t	data.parameters.program	active-response/bin/remove-threat.sh

Step 8: Incident Visibility in TheHive

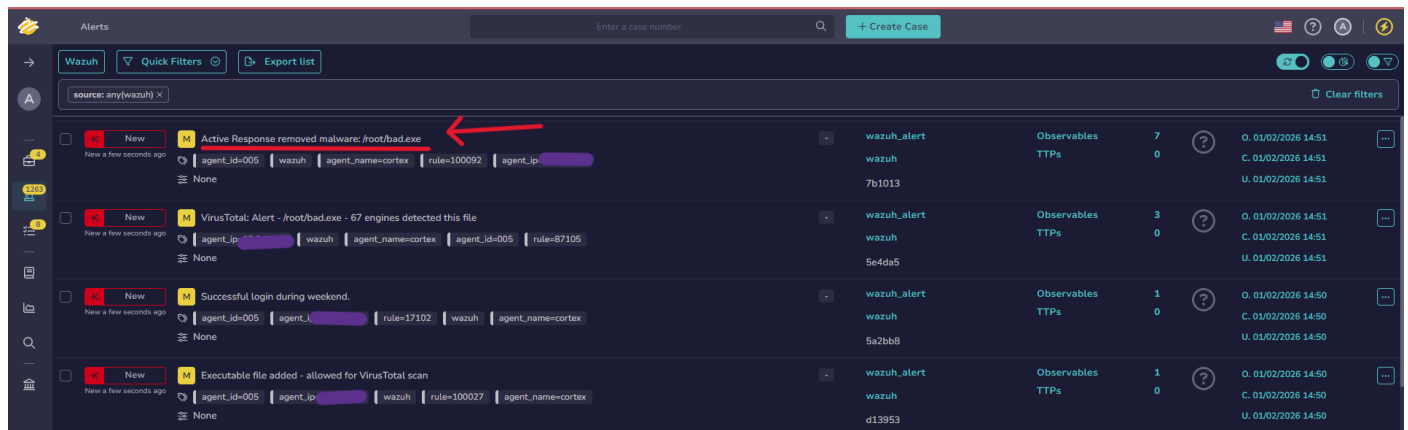
Purpose:

Ensure incidents are visible in IR tooling.

Action:

TheHive successfully received:

- Malware detection alert
- Active response confirmation



Project Outcome

- ✓ Malware detected automatically
- ✓ File reputation verified via VirusTotal
- ✓ Malicious file removed instantly
- ✓ SOC visibility maintained
- ✓ Zero manual intervention required

Skills Gained

- Wazuh Active Response configuration
- Writing SOC-grade Bash automation
- VirusTotal API integration
- File Integrity Monitoring (FIM) tuning
- SOC alert correlation
- Incident response automation
- Endpoint security hardening

Conclusion

This project demonstrates a **real-world SOC automation use case** where malware is not only detected but **immediately neutralized**. By integrating Wazuh Active Response with VirusTotal, organizations can significantly reduce response time, analyst fatigue, and security risk.

This approach is scalable, efficient, and aligns with **modern XDR and SOAR practices** used in enterprise security environments.